In [2]:
```
#supress warnings
import warnings
warnings.simplefilter(action='ignore', category=FutureWarning)
import sys
import os
_stderr = sys.stderr
null = open(os.devnull,'wb')
```

In [4]:
```
import os
os.getcwd()
work_dir = 'res'
```

In [61]:
```
import scanpy as sc
#set some figure parameters for nice display inside jupyternotebooks.
%matplotlib inline
sc.settings.set_figure_params(dpi=80, frameon=False, figsize=(5, 5), facecol

#make a directory for to store the processed scRNA-seq data.
if not os.path.exists(os.path.join(work_dir, 'scRNA')):
    os.makedirs(os.path.join(work_dir, 'scRNA'))
```

Read in the scRNA-seq count matrix into AnnData object.

In [62]:
```
adata = sc.read_h5ad(os.path.join(work_dir, "scRNA/P7_rna_tumor.h5ad"))
adata
```

Out[62]:
```
AnnData object with n_obs × n_vars = 6000 × 22099
    obs: 'orig.ident', 'nCount_RNA', 'nFeature_RNA', 'sample_fullname', 'ba
tch', 'sample', 'PBN_sensitivity', 'patient', 'percent.mt', 'nCount_SCT',
'nFeature_SCT', 'S.Score', 'G2M.Score', 'Phase', 'SCT_snn_res.0.4', 'seurat
_clusters', 'predicted.celltype.l1.score', 'predicted.celltype.l1', 'predic
ted.celltype.l2.score', 'predicted.celltype.l2', 'predicted.celltype.l3.sco
re', 'predicted.celltype.l3', 'mapping.score', 'SCT_snn_res.0.2'
    var: 'features'
    obsm: 'X_integrated_dr', 'X_pca', 'X_umap'
    varm: 'PCs'
```

In [22]:
```
adata.var_names_make_unique()
```

In [ ]:
```
adata.obs['seurat_clusters'] = adata.obs['seurat_clusters'].astype(str)
sc.pl.embedding(adata, color = 'seurat_clusters',basis='umap')
```

In [5]:
```
import os
import pycisTopic
import pandas as pd
#set some figure parameters for nice display inside jupyternotebooks.
%matplotlib inline

#make a directory for to store the processed scRNA-seq data.
if not os.path.exists(os.path.join(work_dir, 'scATAC')):
    os.makedirs(os.path.join(work_dir, 'scATAC'))
tmp_dir = '/Users/fyan1/'
```

load scATAC-seq peak counts & Create cisTopic object

In [25]:
```
# load peak count matrix
from pycisTopic.cistopic_class import *
count_matrix = pd.read_csv(os.path.join(work_dir, 'scATAC/p7_peak_counts_sub
count_matrix.head()
```

Out[25]:

	P_S7_AAACGAACACTGATAC.1	P_S7_AAACGAATCCACGAGC.1	P_S7_AAA(
chr1:9776-10668	0	0	
chr1:180622-181573	0	0	
chr1:267558-268466	0	0	
chr1:585750-586648	0	0	
chr1:629483-630392	0	0	

5 rows × 6000 columns

In [29]:
```
count_matrix.shape
```

Out[29]:
```
(159290, 6000)
```

In [30]:
```
# load metadata
atac_meta = pd.read_csv(os.path.join(work_dir, 'scATAC/p7_atac_meta.txt'), s
atac_meta['sample_id'] = 'p7'
atac_meta['PBN_sensitivity'] = atac_meta['PBN_sensitivity'].astype(str)
atac_meta = atac_meta.rename(index=lambda x: x.replace('-', '.'))
atac_meta.head()
```

Out[30]:

	orig.ident	nCount_ATAC	nFeature_ATAC	total	du
P_S7_AAACGAACACTGATAC.1	SeuratProject	17836	13365	51636	
P_S7_AAACGAATCCACGAGC.1	SeuratProject	13741	11193	33955	
P_S7_AAACGAATCCCGAAGC.1	SeuratProject	20859	15002	55675	
P_S7_AAACGAATCGCTAGTA.1	SeuratProject	22007	16312	55583	
P_S7_AAACTCGCAGTAACGG.1	SeuratProject	16598	12834	47984	

5 rows × 64 columns

In [31]:
```
# check the length of intersected barcodes
len(count_matrix.columns.intersection(atac_meta.index))
```

Out[31]: 6000

In [32]:
```
# create cistopic object
path_to_blacklist='/Users/fyan1/pycisTopic/blacklist/hg38-blacklist.v2.bed'
cistopic_obj = create_cistopic_object(fragment_matrix=count_matrix,
                                      path_to_blacklist=path_to_blacklist)
```

```
2024-06-26 14:59:27,222 cisTopic     INFO     Converting fragment matrix to
sparse matrix
2024-06-26 14:59:34,630 cisTopic     INFO     Removing blacklisted regions
2024-06-26 14:59:35,008 cisTopic     INFO     Creating CistopicObject
2024-06-26 14:59:35,886 cisTopic     INFO     Done!
```

In [33]:
```
print(cistopic_obj)
```

```
CistopicObject from project cisTopic with n_cells × n_regions = 6000 × 15801
1
```

In [34]:
```
# Adding cell information
cistopic_obj.add_cell_data(atac_meta)
```

```
Columns ['sample_id'] will be overwritten
```

In [35]:
```
cistopic_obj.cell_data.head()
```

Out[35]:

	cisTopic_nr_acc	cisTopic_log_nr_frag	cisTop
P_S7_AAACGAACACTGATAC.1___cisTopic	13247	4.247089	
P_S7_AAACGAATCCACGAGC.1___cisTopic	11108	4.134591	
P_S7_AAACGAATCCCGAAGC.1___cisTopic	14887	4.315928	
P_S7_AAACGAATCGCTAGTA.1___cisTopic	16190	4.339372	
P_S7_AAACTCGCAGTAACGG.1___cisTopic	12720	4.215955	

5 rows × 68 columns

In [36]:
```
pickle.dump(cistopic_obj,
            open(os.path.join(work_dir, 'scATAC/cistopic_obj.pkl'), 'wb'))
```

In [54]:
```
import pickle
cistopic_obj = pickle.load(open(os.path.join(work_dir, 'scATAC/cistopic_obj.
from pycisTopic.cistopic_class import *
models=run_cgs_models(cistopic_obj,
                    n_topics=[2,4,10,16,32,48],
                    n_cpu=8,
                    n_iter=500,
                    random_state=555,
                    alpha=50,
                    alpha_by_topic=True,
                    eta=0.1,
                    eta_by_topic=False,
                    save_path=None,
                    _temp_dir = os.path.join(tmp_dir + 'ray_spill'))
```

```
(run_cgs_model pid=61112) 2024-06-27 10:58:33,756 cisTopic     INFO     Runn
ing model with 2 topics
(run_cgs_model pid=61112) 2024-06-27 11:17:41,689 cisTopic     INFO     Mode
l with 2 topics done!
(run_cgs_model pid=61113) 2024-06-27 11:17:41,689 cisTopic     INFO     Mode
l with 2 topics done!
(run_cgs_model pid=61112) 2024-06-27 11:17:41,722 cisTopic     INFO     Runn
ing model with 10 topics
(run_cgs_model pid=61113) 2024-06-27 11:23:37,296 cisTopic     INFO     Mode
l with 4 topics done!
(run_cgs_model pid=61113) 2024-06-27 11:23:37,326 cisTopic     INFO     Runn
ing model with 16 topics
(run_cgs_model pid=61112) 2024-06-27 11:49:51,254 cisTopic     INFO     Mode
l with 10 topics done!
(run_cgs_model pid=61112) 2024-06-27 11:49:51,329 cisTopic     INFO     Runn
ing model with 32 topics
(run_cgs_model pid=61113) 2024-06-27 11:58:24,516 cisTopic     INFO     Mode
l with 16 topics done!
(run_cgs_model pid=61113) 2024-06-27 11:58:24,545 cisTopic     INFO     Runn
ing model with 48 topics
(run_cgs_model pid=61112) 2024-06-27 12:47:13,168 cisTopic     INFO     Mode
l with 32 topics done!
(run_cgs_model pid=61113) 2024-06-27 13:19:57,356 cisTopic     INFO     Mode
l with 48 topics done!
```

In [55]:

```
# save results
if not os.path.exists(os.path.join(work_dir, 'scATAC/models')):
    os.makedirs(os.path.join(work_dir, 'scATAC/models'))

pickle.dump(models,
            open(os.path.join(work_dir, 'scATAC/models/multiome_models_500_i
```

Analyze models

We will make use of four quality metrics to select the model with the optimal amount of topics:

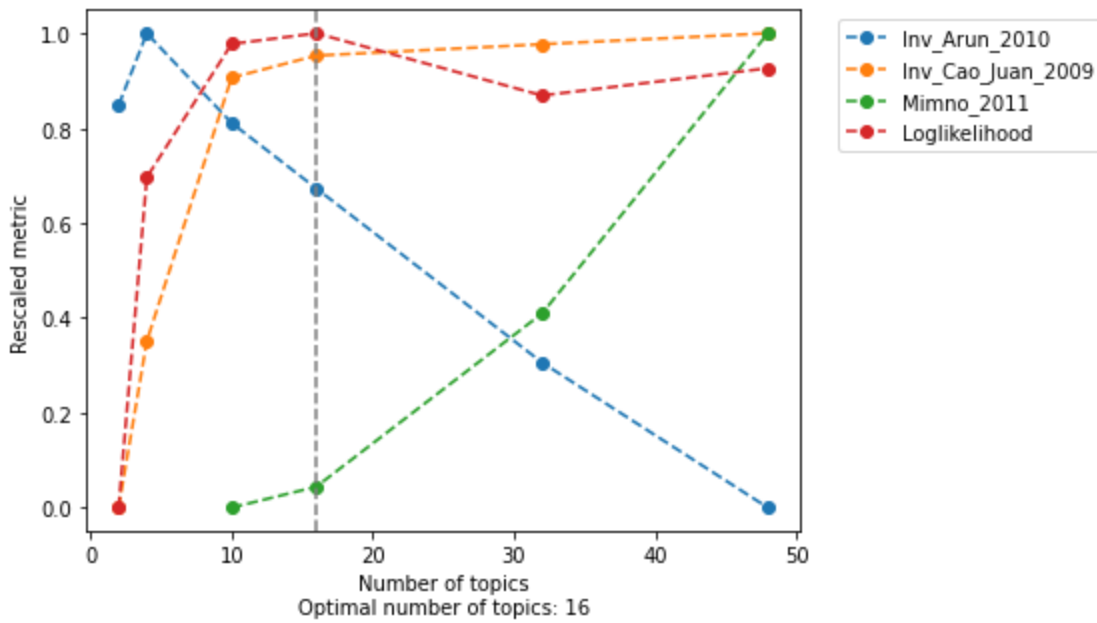
1. Arun *et al.* 2010 (The better the model, the lower the metric.)
2. Cao & Juan *et al.* 2009 (The better the model, the lower the metric.)
3. Mimno *et al.* 2011 (The better the model, the higher coherence.)
4. Log likelihood (The better the model, the higher the log-likelihood.)

For more information on these metrics see publications (linked above) and the read the docs page of pycisTopic.

In [6]:

```
import pickle
import os
work_dir = 'P7_res'
from pycisTopic.cistopic_class import *
models = pickle.load(open(os.path.join(work_dir, 'scATAC/models/multiome_moc
cistopic_obj = pickle.load(open(os.path.join(work_dir, 'scATAC/cistopic_obj.
from pycisTopic.lda_models import *
model = evaluate_models(models,
```

```
                                select_model=16, # select optimal topic number
                                return_model=True,
                                metrics=['Arun_2010','Cao_Juan_2009', 'Minmo_2011', '
                                plot_metrics=False)
```



The metrics seem to stabelise with a model using 16 topics, so let's choose that model.

In [7]:
```
cistopic_obj.add_LDA_model(model)
pickle.dump(cistopic_obj,
            open(os.path.join(work_dir, 'scATAC/cistopic_obj.pkl'), 'wb'))
```

Inferring candidate enhancer regions

Next we will infer candidate enhancer regions by:

1. binarization of region-topic probabilites.
2. calculation differentially accessibile regions (DARs) per cell type.

These regions will be used as input for the next step, pycistarget, in which we will look which motifs are enriched in these regions.

First we will binarize the topics using the otsu method and by taking the top 3k regions per topic.

In [8]:
```
from pycisTopic.topic_binarization import *
region_bin_topics_otsu = binarize_topics(cistopic_obj, method='otsu')
region_bin_topics_top3k = binarize_topics(cistopic_obj, method='ntop', ntop
```

```
<Figure size 460.8x345.6 with 0 Axes>
<Figure size 460.8x345.6 with 0 Axes>
```

Next we will calculate DARs per PBN_sensitivity

In [10]:
```
#!/Users/fyan1/miniconda3/envs/scenicplus/bin/pip install numba==0.57.0
# numba.__version__
```

In [3]:
```
import pycisTopic
pycisTopic.__version__
```

Out[3]:
```
'1.0.3.dev17+g0b5f4d1'
```

In [ ]:
```
from pycisTopic.diff_features import *
imputed_acc_obj = impute_accessibility(cistopic_obj, selected_cells=None,
                                       selected_regions=None, scale_factor=1
normalized_imputed_acc_obj = normalize_scores(imputed_acc_obj, scale_factor=
variable_regions = find_highly_variable_features(normalized_imputed_acc_obj,
markers_dict = find_diff_features(cistopic_obj, imputed_acc_obj, variable='s
                                  var_features=variable_regions, split_patte
```

In [10]:
```
print("Number of DARs found:")
print("---------------------")
for x in markers_dict:
    print(f"  {x}: {len(markers_dict[x])}")
```

```
Number of DARs found:
---------------------
  P_R7: 16151
  P_S7: 5450
```

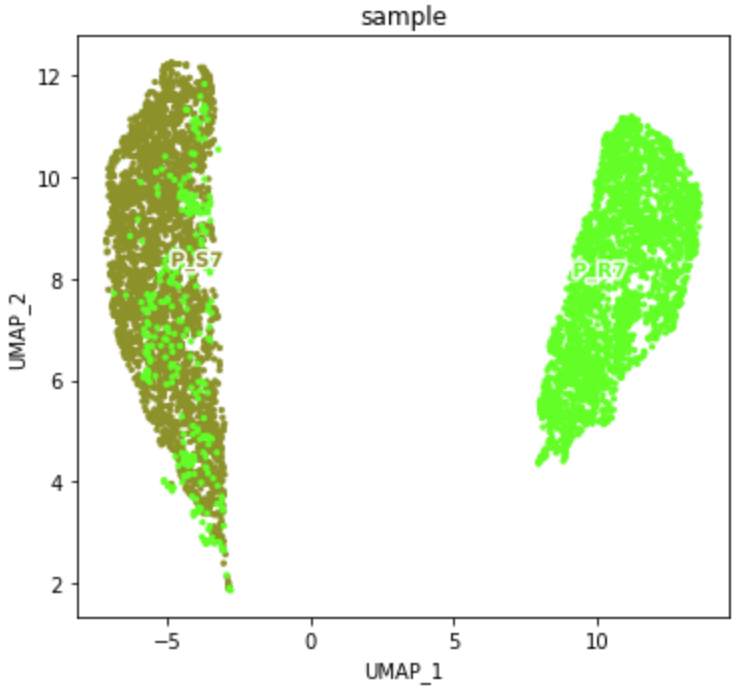
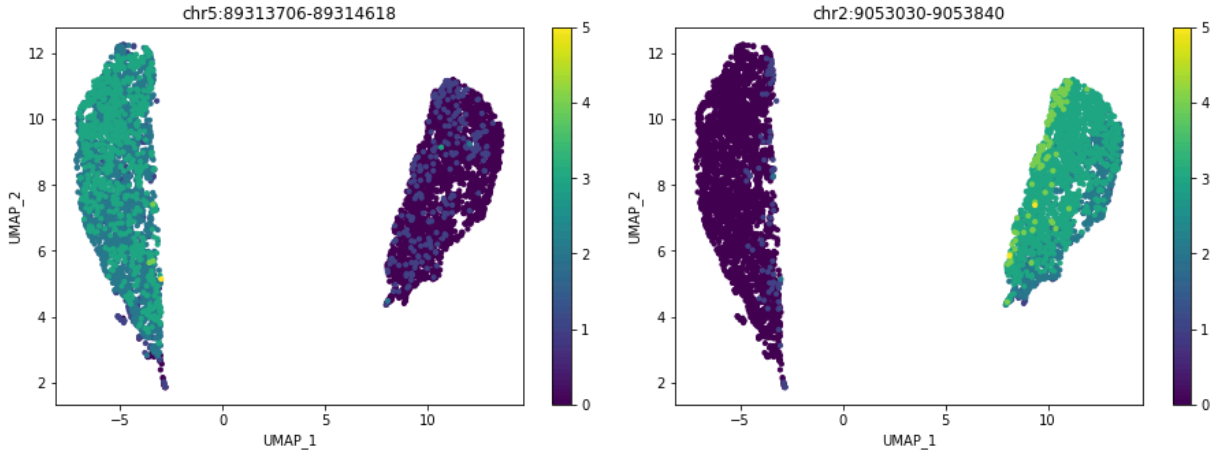
In [13]:
```
# Save results
if not os.path.exists(os.path.join(work_dir, 'scATAC/candidate_enhancers')):
    os.makedirs(os.path.join(work_dir, 'scATAC/candidate_enhancers'))
import pickle
pickle.dump(region_bin_topics_otsu, open(os.path.join(work_dir, 'scATAC/canc
pickle.dump(region_bin_topics_top3k, open(os.path.join(work_dir, 'scATAC/car
pickle.dump(markers_dict, open(os.path.join(work_dir, 'scATAC/candidate_enha
```

In [23]:
```
from pycisTopic.clust_vis import *
run_umap(cistopic_obj,
                 target  = 'cell', scale=False)
```

```
2024-06-26 17:46:02,360 cisTopic     INFO     Running UMAP
```

In [ ]:
```
plot_topic(cistopic_obj, reduction_name = 'UMAP',num_columns=4)
```

In [25]:
```
plot_metadata(cistopic_obj,
                 reduction_name='UMAP',
                 variables=['sample'],
                 target='cell', num_columns=2,
                 text_size=10,
                 dot_size=5,
              #save= 'P8_atac_umap.pdf',
                 figsize=(10,5))
```



In [26]:
```
from pycisTopic.clust_vis import plot_imputed_features
plot_imputed_features(
    cistopic_obj,
    reduction_name='UMAP',
    imputed_data=imputed_acc_obj,
    features=[markers_dict[x].index.tolist()[0] for x in ['P_S7', 'P_R7']],
    scale=False,
    num_columns=4
)
```